Statistics & Probability

ClipSAT Printable Practice Suite (clipsat.org) • Topic 4.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Exponential equations

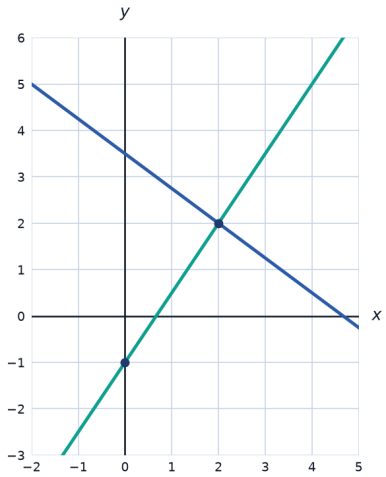
For the model $f(x) = 5x + 4$ in exponential equations, evaluate $f(3)$.

2 [Foundational] • Target: Exponential equations

For the model $f(x) = 5x + 4$ in exponential equations, evaluate $f(4)$.

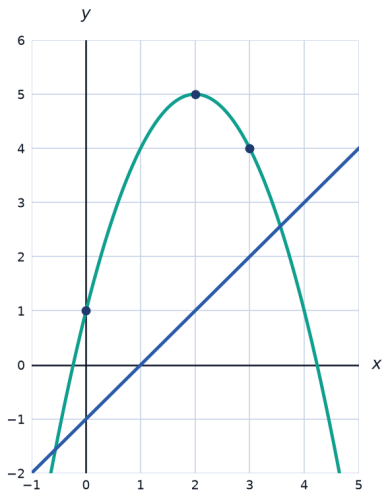
3 [Foundational] • Target: Exponential equations

For the model $f(x) = 5x + 4$ in exponential equations, evaluate $f(5)$.



4 [Foundational] • Target: The natural exponent

For the model $f(x) = 6x + 5$ in the natural exponent, evaluate $f(2)$.



5 [Foundational] • Target: The natural exponent

Find the positive solutions to the quadratic equation $x^2 - 6x + 5 = 0$.

© ClipSAT — clipsat.org | All Rights Reserved

6 [Foundational] • Target: The natural exponent

A data set recorded for the natural exponent consists of the four values $\{16, 20, 24, 30\}$. What is the arithmetic mean? Show your complete algebraic work.

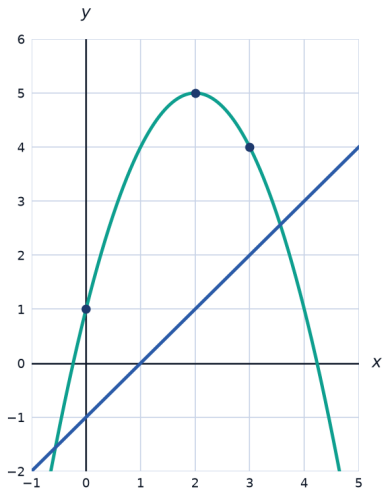
7 [Foundational] • Target: The natural exponent

For the model $f(x) = 6x + 5$ in the natural exponent, evaluate $f(3)$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: The natural exponent

Find the positive solutions to the quadratic equation $x^2 - 8x + 12 = 0$.



9 [Intermediate] • Target: The natural exponent

For the model $f(x) = 6x + 5$ in the natural exponent, evaluate $f(4)$.

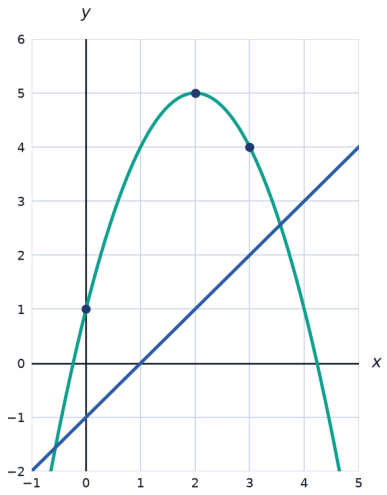
10 [Intermediate] • Target: The natural exponent

Find the positive solutions to the quadratic equation $x^2 - 10x + 21 = 0$.

© ClipSAT — clipsat.org | All Rights Reserved

11 [Intermediate] • Target: The natural exponent

For the model $f(x) = 6x + 5$ in the natural exponent, evaluate $f(5)$.

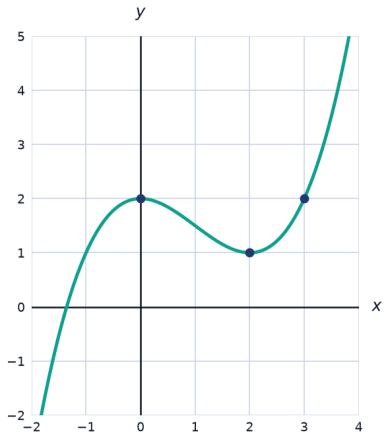


12 [Intermediate] • Target: The natural exponent

Find the positive solutions to the quadratic equation $x^2 - 12x + 32 = 0$.

13 [Intermediate] • Target: Logarithms

For the model $f(x) = 2x + 6$ in logarithms, evaluate $f(2)$.



14 [Intermediate] • Target: Logarithms

Find the positive solutions to the quadratic equation $x^2 - 7x + 6 = 0$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Logarithms • Cross-Lesson: 3.01 Geometry & Trigonometry

A data set recorded for logarithms consists of the four values $\{17, 21, 25, 31\}$. What is the arithmetic mean? Show your complete algebraic work.

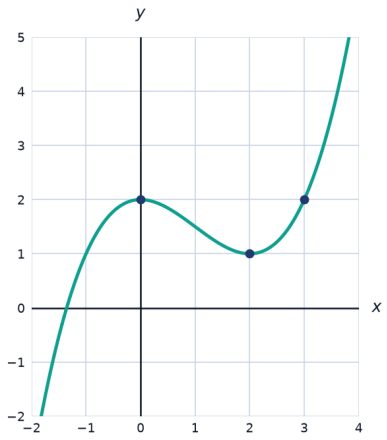
© ClipSAT — clipsat.org | All Rights Reserved

16 [Challenging] • Target: Logarithms • Cross-Lesson: 5.01 Calculus

For the model $f(x) = 2x + 6$ in logarithms, evaluate $f(3)$.

17 [Challenging] • Target: Logarithms • Cross-Lesson: 6.01 AI SL — Specific Topics

Find the positive solutions to the quadratic equation $x^2 - 9x + 14 = 0$.



18 [Challenging] • Target: Logarithms • Cross-Lesson: 1.01 Number & Algebra

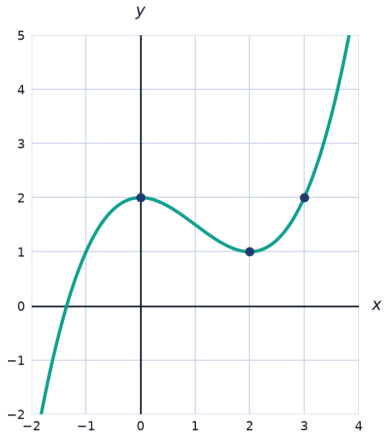
For the model $f(x) = 2x + 6$ in logarithms, evaluate $f(4)$.

19 [Challenging] • Target: Logarithms • Cross-Lesson: 2.01 Functions

Find the positive solutions to the quadratic equation $x^2 - 11x + 24 = 0$.

20 [Challenging] • Target: Logarithms • Cross-Lesson: 3.01 Geometry & Trigonometry

For the model $f(x) = 2x + 6$ in logarithms, evaluate $f(5)$.

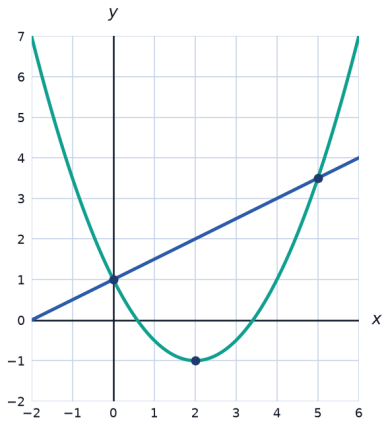


Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

© ClipSAT — clipsat.org | All Rights Reserved

21 [Foundational] • Target: Logic

For the model $f(x) = 5x + 3$ in logic, evaluate $f(2)$.



a. 13

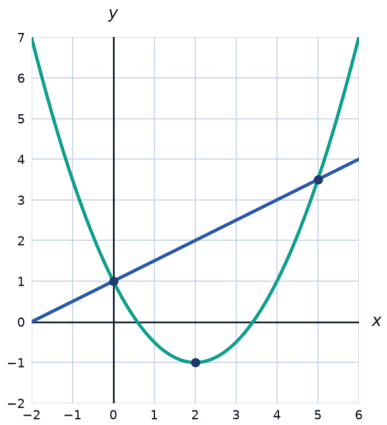
b. 18

c. 10

d. 17

22 [Foundational] • Target: Logic

Find the positive solutions to the quadratic equation $x^2 - 6x + 8 = 0$.



a. $x = 2$ and $x = 5$

b. $x = 2$ and $x = 4$

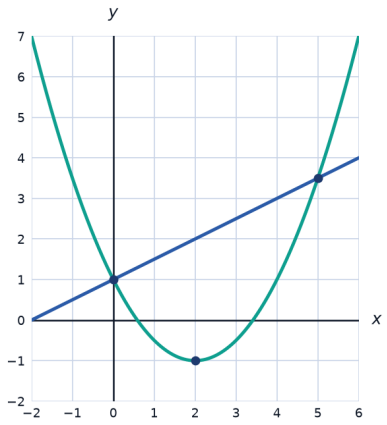
c. $x = -2$ and $x = -4$

d. $x = 6$ and $x = 8$

© ClipSAT — clipsat.org | All Rights Reserved

23 [Foundational] • Target: Logic

For the model $f(x) = 5x + 3$ in logic, evaluate $f(5)$.



a. 28

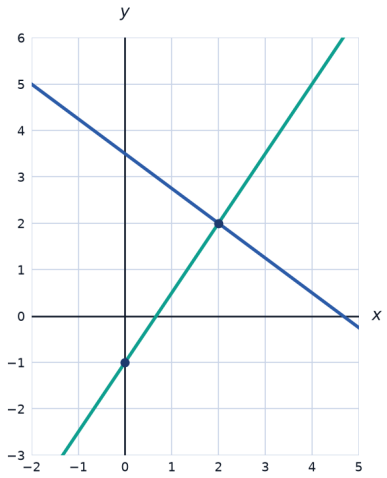
b. 33

c. 25

d. 32

24 [Foundational] • Target: Proof by deduction

For the model $f(x) = 6x + 4$ in proof by deduction, evaluate $f(2)$.



a. 16

b. 22

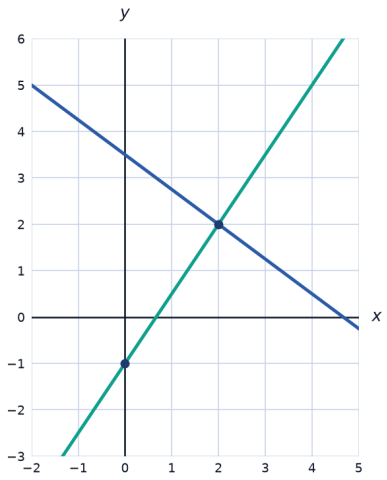
c. 12

d. 20

© ClipSAT — clipsat.org | All Rights Reserved

25 [Foundational] • Target: Proof by deduction

Find the positive solutions to the quadratic equation $x^2 - 7x + 10 = 0$.



a. $x = 2$ and $x = 6$

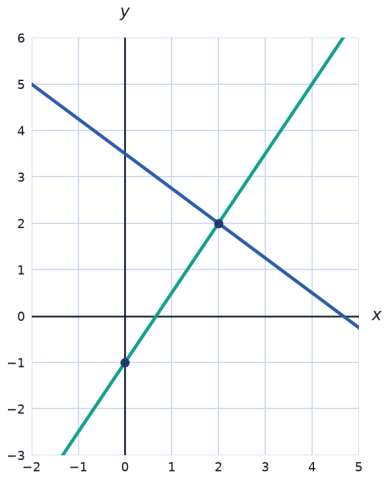
b. $x = 2$ and $x = 5$

c. $x = -2$ and $x = -5$

d. $x = 7$ and $x = 10$

26 [Foundational] • Target: Proof by deduction

For the model $f(x) = 6x + 4$ in proof by deduction, evaluate $f(5)$.



a. 34

b. 40

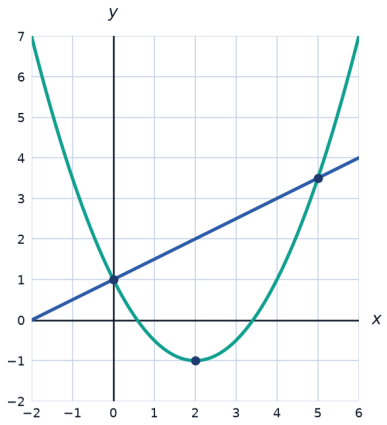
c. 30

d. 38

© ClipSAT — clipsat.org | All Rights Reserved

27 [Foundational] • Target: Applications of exponential functions

For the model $f(x) = 4x + 3$ in applications of exponential functions, evaluate $f(2)$.



a. 11

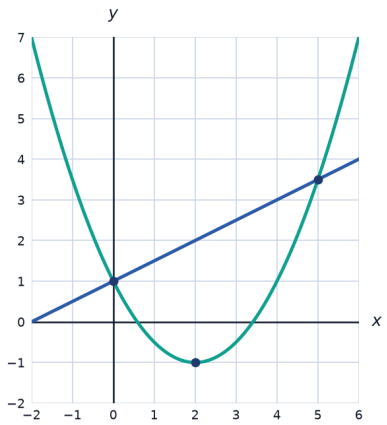
b. 15

c. 8

d. 15

28 [Foundational] • Target: Applications of exponential functions

For the model $f(x) = 4x + 3$ in applications of exponential functions, evaluate $f(5)$.



a. 23

b. 27

c. 20

d. 27

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

© ClipSAT — clipsat.org | All Rights Reserved

29 [Intermediate] • Target: Logic

Find the positive solutions to the quadratic equation $x^2 - 4x + 3 = 0$.

a. $x = 1$ and $x = 4$

b. $x = 1$ and $x = 3$

c. $x = -1$ and $x = -3$

d. $x = 4$ and $x = 3$

30 [Intermediate] • Target: Logic

A data set recorded for logic consists of the four values $\{11, 15, 19, 25\}$. What is the arithmetic mean?

a. 15.5

b. 16.5

c. 17.5

d. 19.5

31 [Intermediate] • Target: Logic

For the model $f(x) = 5x + 3$ in logic, evaluate $f(3)$.

a. 18

b. 23

c. 15

d. 22

32 [Intermediate] • Target: Logic

For the model $f(x) = 5x + 3$ in logic, evaluate $f(4)$.

a. 23

b. 28

c. 20

d. 27

33 [Intermediate] • Target: Logic

Find the positive solutions to the quadratic equation $x^2 - 8x + 15 = 0$.

a. $x = 3$ and $x = 6$

b. $x = 3$ and $x = 5$

c. $x = -3$ and $x = -5$

d. $x = 8$ and $x = 15$

34 [Intermediate] • Target: Logic

Find the positive solutions to the quadratic equation $x^2 - 10x + 24 = 0$.

a. $x = 4$ and $x = 7$

b. $x = 4$ and $x = 6$

c. $x = -4$ and $x = -6$

d. $x = 10$ and $x = 24$

© ClipSAT — clipsat.org | All Rights Reserved

35 [Intermediate] • Target: Proof by deduction

Find the positive solutions to the quadratic equation $x^2 - 5x + 4 = 0$.

a. $x = 1$ and $x = 5$

b. $x = 1$ and $x = 4$

c. $x = -1$ and $x = -4$

d. $x = 5$ and $x = 4$

36 [Intermediate] • Target: Proof by deduction

A data set recorded for proof by deduction consists of the four values $\{12, 16, 20, 26\}$. What is the arithmetic mean?

a. 16.5

b. 17.5

c. 18.5

d. 20.5

37 [Intermediate] • Target: Proof by deduction

For the model $f(x) = 6x + 4$ in proof by deduction, evaluate $f(3)$.

a. 22

b. 28

c. 18

d. 26

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Proof by deduction • Cross-Lesson: 1.01 Number & Algebra

Find the positive solutions to the quadratic equation $x^2 - 9x + 18 = 0$.

a. $x = 3$ and $x = 7$

b. $x = 3$ and $x = 6$

c. $x = -3$ and $x = -6$

d. $x = 9$ and $x = 18$

39 [Challenging] • Target: Proof by deduction • Cross-Lesson: 2.01 Functions

Find the positive solutions to the quadratic equation $x^2 - 11x + 28 = 0$.

a. $x = 4$ and $x = 8$

b. $x = 4$ and $x = 7$

c. $x = -4$ and $x = -7$

d. $x = 11$ and $x = 28$

© ClipSAT — clipsat.org | All Rights Reserved

40 [Challenging] • Target: Applications of exponential functions • Cross-Lesson: 3.01 Geometry & Trigonometry

A data set recorded for applications of exponential functions consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean?

a. 18.5

b. 19.5

c. 20.5

d. 22.5

41 [Challenging] • Target: Applications of exponential functions • Cross-Lesson: 5.01 Calculus

For the model $f(x) = 4x + 3$ in applications of exponential functions, evaluate $f(3)$.

a. 15

b. 19

c. 12

d. 19

42 [Challenging] • Target: Applications of exponential functions • Cross-Lesson: 6.01 AI SL — Specific Topics

For the model $f(x) = 4x + 3$ in applications of exponential functions, evaluate $f(4)$.

a. 19

b. 23

c. 16

d. 23

43 [Challenging] • Target: Exponential equations • Cross-Lesson: 1.01 Number & Algebra

A data set recorded for exponential equations consists of the four values $\{15, 19, 23, 29\}$. What is the arithmetic mean?

a. 19.5

b. 20.5

c. 21.5

d. 23.5

44 [Challenging] • Target: The natural exponent • Cross-Lesson: 2.01 Functions

A data set recorded for the natural exponent consists of the four values $\{16, 20, 24, 30\}$. What is the arithmetic mean?

a. 20.5

b. 21.5

c. 22.5

d. 24.5

© ClipSAT — clipsat.org | All Rights Reserved

45 [Challenging] • Target: Logarithms • Cross-Lesson: 3.01 Geometry & Trigonometry

A data set recorded for logarithms consists of the four values $\{17, 21, 25, 31\}$. What is the arithmetic mean?

a. 21.5

b. 22.5

c. 23.5

d. 25.5

© ClipSAT — clipsat.org | All Rights Reserved